CS & IT

ENGINEERING

COMPILER DESIGN

PARSING

Lecture No.- 09

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Recap of Previous Lecture

Topic

?? {Operator Precedence Parsing}

✓ LR Parsing

Topics to be Covered

Topic ?? Operator Precedence Parsing

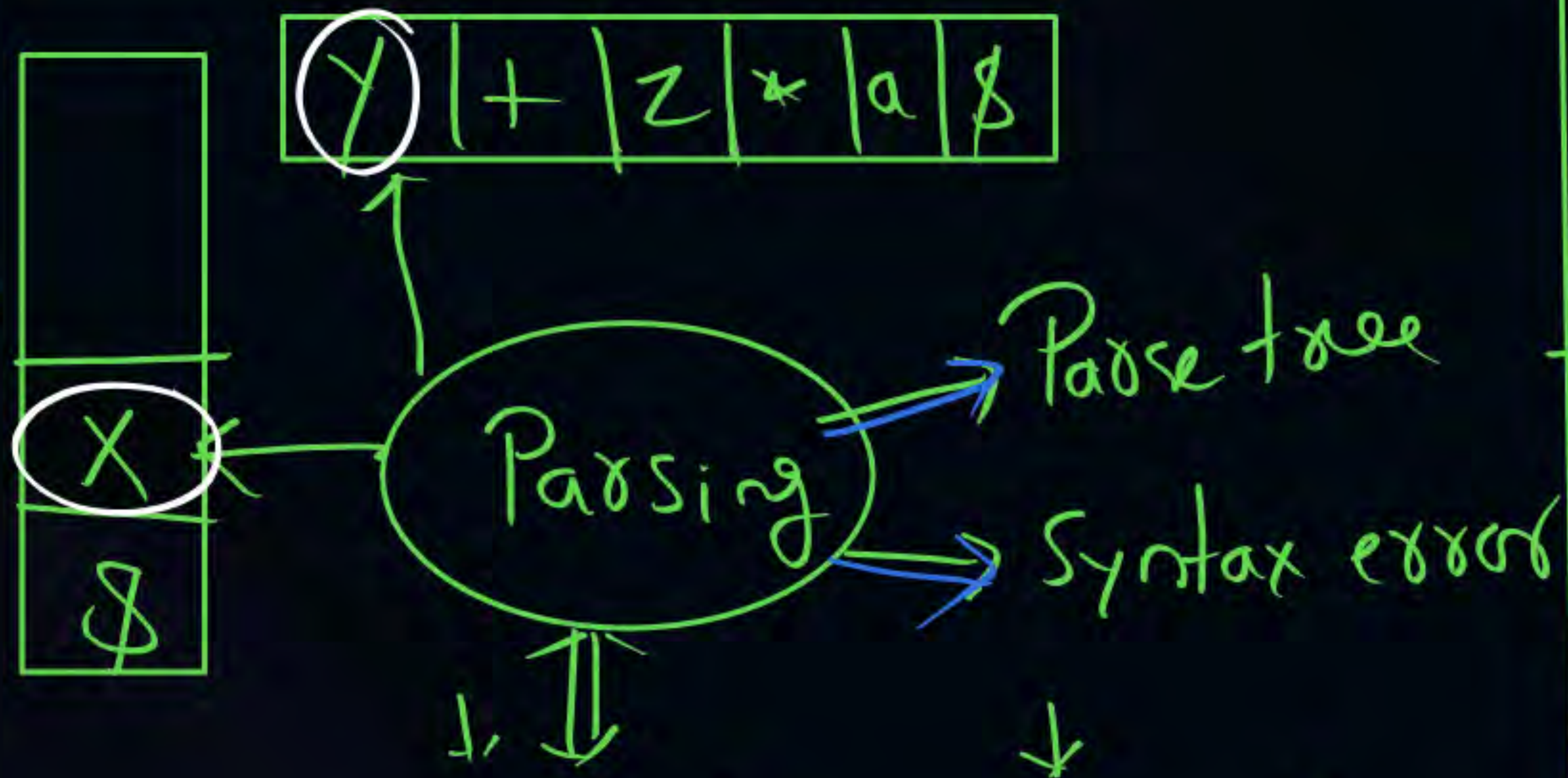
Topic ?? Topdown Parsing

Topic ?? LL(1) Parsing

Operator Precedencde Parsing

Time Complexity:

$O(n)$



$E \to E+E$

$E \to E*E$

$E \to a$

	a	*	+	$
a		$\gtrdot$	$\gtrdot$	$\gtrdot$
*	$\lessdot$	$\gtrdot$	$\gtrdot$	$\gtrdot$
+	$\lessdot$	$\lessdot$	$\gtrdot$	$\gtrdot$
$	$\lessdot$	$\lessdot$	$\lessdot$	accept

1. $\left.\begin{array}{l} X \lessdot Y \\ X \doteq Y \end{array}\right\}$ shift Y

2. $X \gtrdot Y$ } Reduce

3. $T[\$,\$]$ = accept

4. Blank = Syntax error

	Stack		input	Action
(1)	| $ |	<·	a+a*a$	Shift
(2)	| $ | a |	·>	+a*a$	Reduce
(3)	| $ | E |	<·	+a*a$	Shift
(4)	| $ | E | +	<·	a*a$	Shift
(5)	| $ | E | + | a	·>	*a$	Reduce
(6)	| $ | E | + | E	<·	*a$	Shift
(7)	| $ | E | + | E | *	<·	a $	Shift
(8)	| $ | E | + | E | * | a	·>	$	Reduce
(9)	| $ | E | + | E | * | E		$	

9	\$ | E | + | E | * | E	$\cdot >$	\$	Reduce
10	\$ | E | + | E	$\cdot >$	\$	Reduce
11	\$ | E		\$	accept

$E \rightarrow E * T \mid T$

$T \rightarrow T + F \mid F$

$F \rightarrow a$

$+ \Rightarrow$ L.A

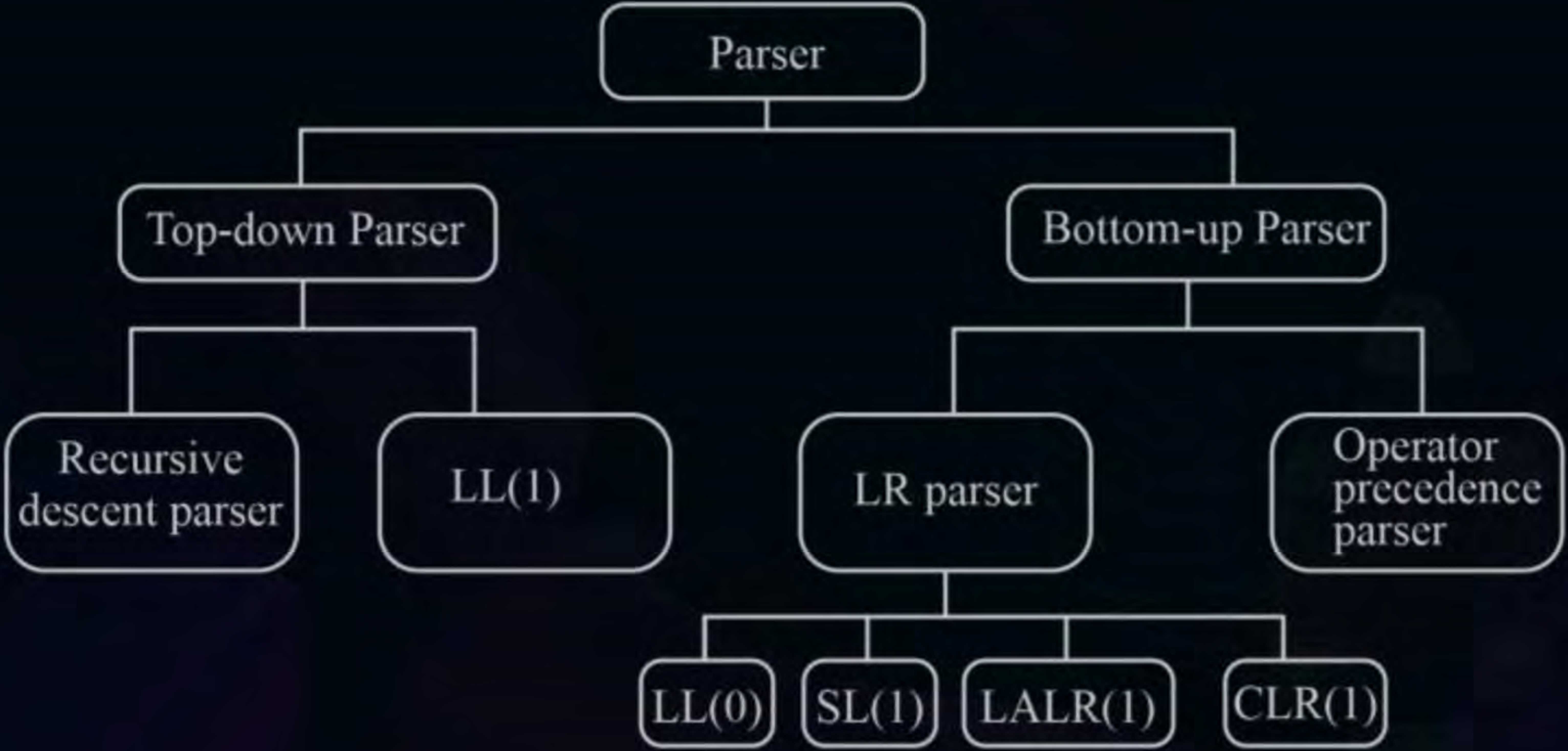
$* \Rightarrow$ L.A

	a	+	*	$
a		$\cdot>$	$\cdot>$	$\cdot>$
+	$<\cdot$	$\cdot>$	$\cdot>$	$\cdot>$
*	$<\cdot$	$<\cdot$	$\cdot>$	$\cdot>$
$	$<\cdot$	$<\cdot$	$<\cdot$	accept

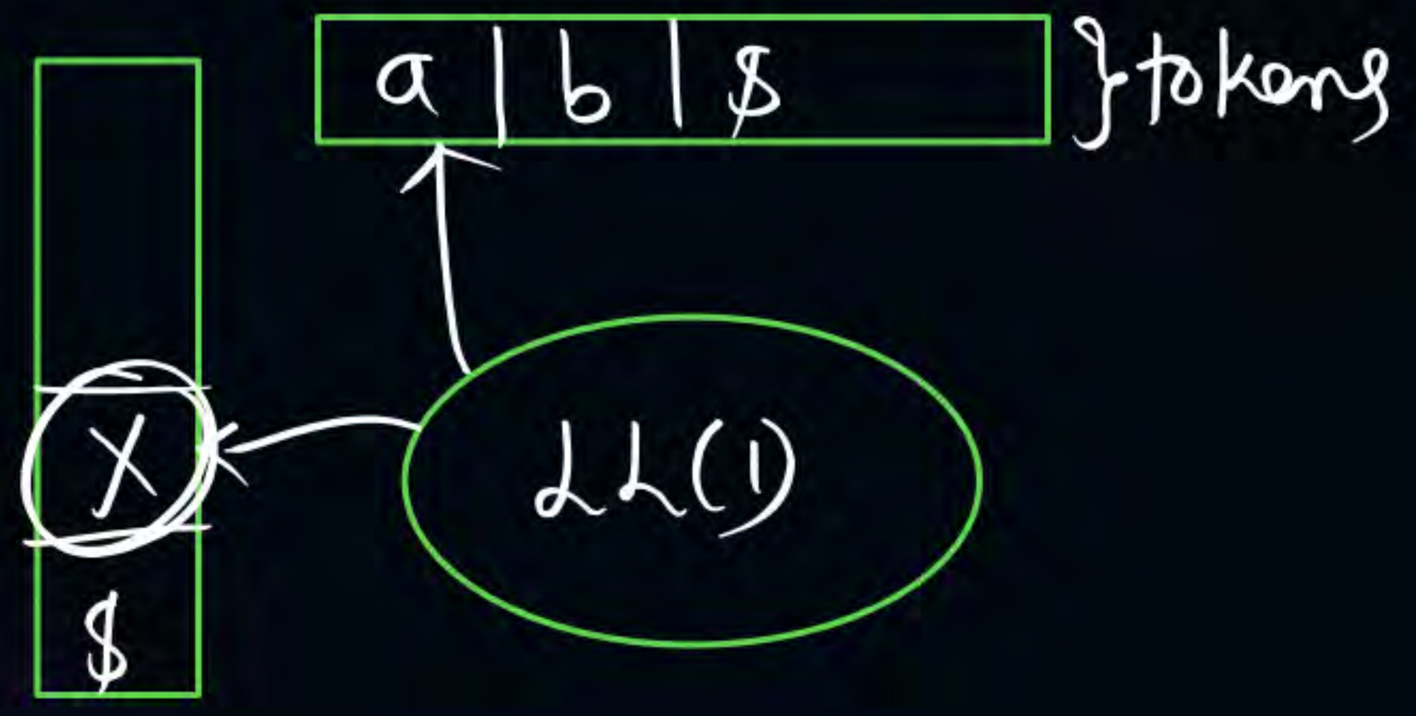
$ is less over all

Id is higher

Topic : PARSER



Topic : LL(1) Parsing



	a	b	$
E			
E'			

LL(1) Parsing table

Topic : LL(1) Parsing

$E' \rightarrow +TE' \mid \epsilon$

Topic : LL(1) Parsing

Construct LL(1) Parsing table for Given grammar

$\{a\}$
First

$S \to aABC$
$A \to a \mid bb$
$B \to a \mid \epsilon$
$C \to b \mid \epsilon$

LL(1) Grammar

Row = N.T
Column = First (R.H.S)

$[S, a] = S \to aABC$

First(a) = {a}

$A \to a$
$A \to bb$
$B \to a$

$\boxed{B \to \epsilon}$

column = First (ε) = {ε}

Follow(B) = {b, $}

$C \to \epsilon$

First(ε) = {ε}
Follow(C) = {$}

	a	b	$
S	$S \to aABC$		
A	$A \to a$	$A \to bb$	
B	$B \to a$	$B \to \epsilon$	$B \to \epsilon$
C		$C \to b$	$C \to \epsilon$

LL(1) Parsing Table

(1) Construct LL(1) Parsing table for given grammar

$S \to aAbB$

$S \to bAaB$

$S \to \epsilon$

$A \to S$

$B \to S$

not LL(1) grammar

$\boxed{S \to \epsilon}$

$First(\epsilon) = \{\epsilon\}$

$Follow(S) = \{a, b, \$\}$

$A \to S$

$First(S) = \{a, b, \epsilon\}$

$Follow(A) = \{a, b\}$

$B \to S$

$First(S) = \{a, b, \epsilon\}$

$Follow(B) = \{a, b, \$\}$

	a 1	b 2	\$ 3
S	$S \to aAbB$, $S \to \epsilon$	$S \to bAaB$, $S \to \epsilon$	$S \to \epsilon$
A	$A \to S$ ✓	$A \to S$ ✓	
B	$B \to S$ ✓	$B \to S$ ✓	$B \to S$

(Q) Construct LL(1) Parsing Table for given grammar

First(AaAb) = {c, a}

$S \to AaAb$

First(BbBa) = {d, b}

$S \to BbBa$

$A \to c \mid \epsilon$

$B \to d \mid \epsilon$

LL(1) grammar

$A \to \epsilon$

First(ϵ) = {ϵ}

Follow(A) = {a, b}

$\boxed{B \to \epsilon}$

First(ϵ) = {ϵ}

Follow(B) = {a, b}

	a	b	c	d	$
S	$S \to AaAb$	$S \to BbBa$	$S \to AaAb$	$S \to BbBa$	
A	$A \to \epsilon$	$A \to \epsilon$	$A \to c$		
B	$B \to \epsilon$	$B \to \epsilon$		$B \to d$	

{LL(1) Parsing Table}

LL(1) Grammar

LL(1) Detection

$\{E \to E + T\}$

$\{S \to ab \mid ac \mid ad\}$

$\{S \to AB,\ A \to a,\ B \to b\}$ ✓

1. Ambiguous Grammar is not LL(1) grammar
2. Left Recursive grammar is not LL(1) grammar
3. Common prefix Grammar is not LL(1) grammar
4. Single production grammar is LL(1) grammar.

(5) $A \rightarrow \alpha_1 \mid \alpha_2 \mid \cdots \mid \alpha_n$

$\left.\text{First}(\alpha_1), \text{First}(\alpha_2) \cdots \text{First}(\alpha_n)\right\}$ If any two contain common element then not LL(1)

(6) $A \rightarrow \alpha \mid \epsilon$

$\left.\text{First}(\alpha) \cap \text{Follow}(A) = \phi\right\}$ LL(1)

① $S \to aSbS \mid bSaS \mid \epsilon$ } not LL(1)

a b Follow(S)

{a, b, \$}

(2) $S \to aABb$

$A \to c \mid \epsilon$

(c) | Follow(A) = {b, d}

$B \to d \mid \epsilon$

(d) || Follow(B) = {b}

LL(1) grammar

③ $S \rightarrow aSA$

$A \rightarrow c \mid \epsilon$

$\{c\} \mid Follow(A)$

$\{c, \$\}$

not LL(1)

(Q) Which of the following is LL(1) grammar?

(a) $E \to E+E \mid E*E \mid a$ ✗

(b) $E \to E+T \mid T$
$T \to a$ ✗

(c) $S \to ab \mid ac$ ✗

(d) none

Topic : LL(1) Parsing

Construct LL(1) Parsing table for Given grammar

$S \rightarrow aABC$

$A \rightarrow a \mid bb$

$B \rightarrow a \mid \epsilon$

$C \rightarrow b \mid \epsilon$

	First	Follow
S	$\{a\}$	$\{\$\}$
A	$\{a, b\}$	$\{a, b, \$\}$
B	$\{a, \epsilon\}$	$\{b, \$\}$
C	$\{b, \epsilon\}$	$\{\$\}$

THANK - YOU